torch.addcdiv#

https://pytorch.org/docs/stable/generated/torch.addcdiv.html#torch.addcdiv

Description:

Performs the element-wise division of tensor1 by tensor2, multiplies the result by the scalar value and adds it to input. Integer division with addcdiv is no longer supported, and in a future release addcdiv will perform a true division of tensor1 and tensor2. The historic addcdiv behavior can be implemented as (input + value * torch.trunc(tensor1 / tensor2)).to(input.dtype) for integer inputs and as (input + value * tensor1 / tensor2) for float inputs. The future addcdiv behavior is just the latter implementation: (input + value * tensor1 / tensor2), for all dtypes. The shapes of input, tensor1, and tensor2 must be broadcastable. For inputs of type FloatTensor or DoubleTensor, value must be a real number, otherwise an integer.

Function Signature

```
torch.addcdiv(input, tensor1, tensor2, *, value=1, out=None)
→ Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

value (Number, optional) – multiplier for tensor1/tensor2\text{tensor1} / \text{tensor2}tensor1/tensor2 out (Tensor, optional) – the output tensor.

Code Examples

```
>>> t = torch.randn(1, 3)
>>> t1 = torch.randn(3, 1)
>>> t2 = torch.randn(1, 3)
>>> torch.addcdiv(t, t1, t2, value=0.1)
tensor([[-0.2312, -3.6496,  0.1312],
        [-1.0428,  3.4292, -0.1030],
        [-0.5369, -0.9829,  0.0430]])
```

Notes & Warnings

⚠ **WARNING**

Warning Integer division with addcdiv is no longer supported, and in a future release addcdiv will perform a true division of tensor1 and tensor2. The historic addcdiv behavior can be implemented as (input + value * torch.trunc(tensor1 / tensor2)).to(input.dtype) for integer inputs and as (input + value * tensor1 / tensor2) for float inputs. The future addcdiv behavior is just the latter implementation: (input + value * tensor1 / tensor2), for all dtypes.

Detailed Documentation

Documentation

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Integer division with addcdiv is no longer supported, and in a future release addcdiv will perform a true division of tensor1 and tensor2. The historic addcdiv behavior can be implemented as (input + value * torch.trunc(tensor1 / tensor2)).to(input.dtype) for integer inputs and as (input + value * tensor1 / tensor2) for float inputs. The future addcdiv behavior is just the latter implementation: (input + value * tensor1 / tensor2), for all dtypes.

The shapes of input, tensor1, and tensor2 must be broadcastable.

For inputs of type FloatTensor or DoubleTensor, value must be a real number, otherwise an integer.

input (Tensor) – the tensor to be added

tensor1 (Tensor) – the numerator tensor

tensor2 (Tensor) – the denominator tensor

value (Number, optional) – multiplier for tensor1/tensor2\text{tensor1} / \text{tensor2}tensor1/tensor2

out (Tensor, optional) – the output tensor.